Network Port Scanning and Traffic Analysis using Nmap and Wireshark

1. Introduction

In modern computer networks, security plays a crucial role in protecting systems from unauthorized access and cyber threats. One of the fundamental steps in network security assessment is **network reconnaissance**, which involves discovering active devices and identifying open ports.
Port scanning is a technique used to determine which ports on a network are open and what services are running on them. Open ports can act as entry points for attackers if not properly secured.
In this experiment, tools like **Nmap** and **Wireshark** were used to perform network scanning and packet-level analysis. This helped in understanding how devices respond to connection attempts and how firewalls protect network infrastructure.

2. Objective

- To identify active hosts in a local network
- To detect open, closed, and filtered ports
- To understand TCP SYN scanning technique
- To analyze packet-level communication using Wireshark
- To evaluate network security posture

3. Tools Used

- **Nmap**

A powerful open-source tool used for network discovery and security auditing. It helps in scanning hosts, detecting open ports, and identifying services.

- **Wireshark**

A network protocol analyzer used to capture and inspect packets in real time.

- **Kali Linux**

A penetration testing operating system used for performing cybersecurity tasks.

4. Network Information

The system's IP address was identified using:

ip a

Observed Output

inet 192.168.59.129/24

Network Range

192.168.59.0/24

This indicates that the system belongs to a Class C private network with 256 possible IP addresses.

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1e:ef:75 brd ff:ff:ff:ff:ff:ff
    inet 192.168.59.129/24 brd 192.168.59.255 scope global dynamic noprefixroute eth0
       valid_lft 1636sec preferred_lft 1636sec
    inet6 fe80::55f2:9e08:9449:90cf/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
┌──(kali㉿kali)-[~]
└─$ nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.5 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

5. Methodology

Step 1: Installation and Setup

Nmap was verified to be installed and working properly.

Step 2: Detailed Scan

sudo nmap -sS -sV -O 192.168.59.0/24 -oN detailed_scan.txt

Purpose:

Detect service versions

Identify operating systems

Save results for analysis

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -sV -O 192.168.59.0/24 -oN detailed_scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 06:12 -0400
Nmap scan report for 192.168.59.1
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.59.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.59.2
Host is up (0.00070s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE    SERVICE VERSION
53/tcp filtered domain
MAC Address: 00:50:56:E8:40:97 (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 o
Device type: specialized
Running: VMware Player
OS CPE: cpe:/a:vmware:player
OS details: VMware Player virtual NAT device
Network Distance: 1 hop

Nmap scan report for 192.168.59.254
Host is up (0.00038s latency).
All 1000 scanned ports on 192.168.59.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E3:2F:C1 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.59.129
Host is up (0.00018s latency).
All 1000 scanned ports on 192.168.59.129 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops
```

Step 3: Packet Capture using Wireshark

Wireshark was launched

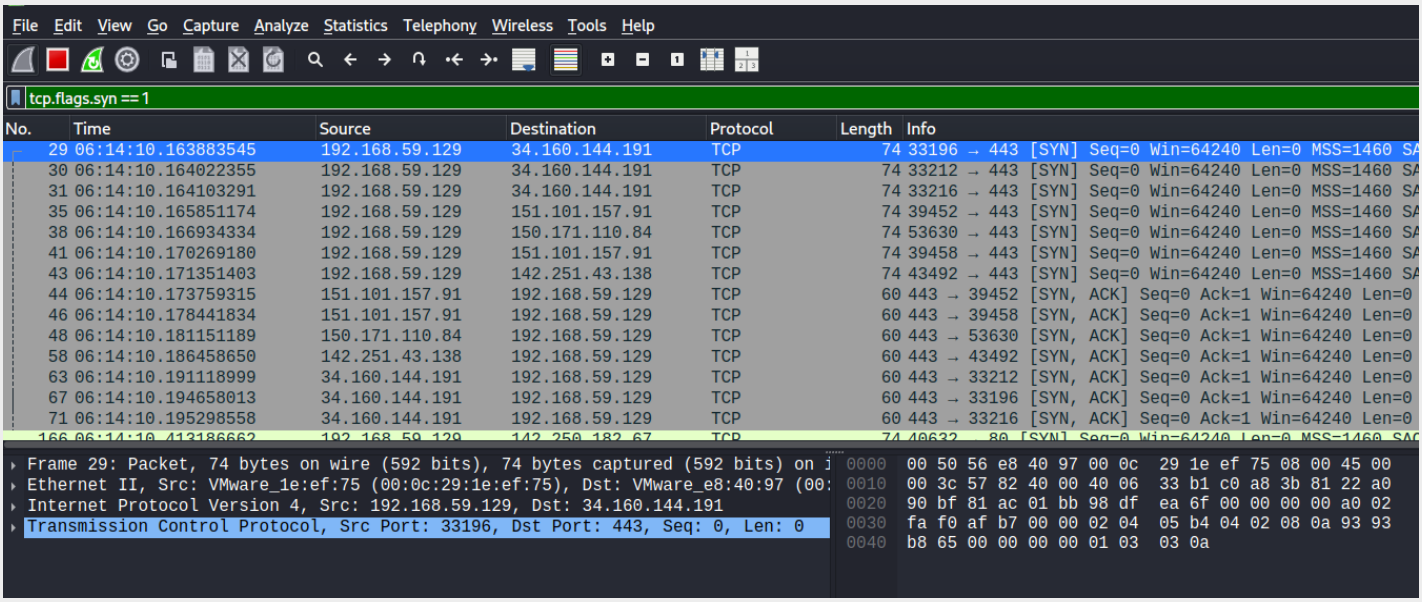
Interface eth0 was selected

Capture was started

Nmap scan was executed

Filter Used:

tcp.flags.syn == 1




```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.59.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 06:21 -0400
Nmap scan report for 192.168.59.1
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.59.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.59.2
Host is up (0.00051s latency).
Not shown: 999 closed tcp ports (reset)
PORT    STATE    SERVICE
53/tcp filtered domain
MAC Address: 00:50:56:E8:40:97 (VMware)

Nmap scan report for 192.168.59.254
Host is up (0.00023s latency).
All 1000 scanned ports on 192.168.59.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E3:2F:C1 (VMware)

Stats: 0:04:04 elapsed; 255 hosts completed (4 up), 1 undergoing SYN Stea
SYN Stealth Scan Timing: About 21.50% done; ETC: 06:39 (0:14:14 remaining
Nmap scan report for 192.168.59.129
Host is up.
```

Results

- **Active Hosts Detected**

IP Address	Status
192.168.59.1	Active
192.168.59.2	Active
192.168.59.254	Active
192.168.59.129	Active

Port Scan Analysis

Host: 192.168.59.1

All 1000 ports filtered

No response from target

Indicates strong firewall protection

Host: 192.168.59.2

Port	State	Service
53	Filtered	DNS

999 ports closed

1 port filtered

Likely VMware NAT device

Host: 192.168.59.254

All ports filtered

No exposed services

Host: 192.168.59.129 (Local Machine)

All ports filtered

System is secure and not exposing services

Wireshark Analysis

Wireshark was used to capture network packets during the Nmap scan.

Observations:

SYN packets were sent from source IP 192.168.59.129

Destination included multiple IPs in the subnet

Majority of packets received no response

Some packets received RST responses

Packet Behavior Analysis

Packet Type	Meaning
SYN	Request to initiate connection
SYN-ACK	Port is open
RST	Port is closed
No response	Port is filtered

Findings:

•No SYN-ACK packets observed → No open ports

•RST packets observed → Closed ports

•No response → Filtered port

Security Assessment

Strengths

No open ports exposed

Firewall protection is active
Minimal attack surface

Potential Risks

DNS service (port 53) detected (filtered)

Misconfiguration could expose services in future

Risk Analysis

Even though no open ports were found, the following risks exist in general:

Open ports can allow unauthorized access

Services may have vulnerabilities

Weak authentication may lead to compromise

Filtered ports indicate security but may also hide services from detection.

Conclusion

The network scan revealed that all devices in the network are well protected, as no open ports were identified. The presence of filtered ports indicates that firewall rules are effectively blocking unauthorized access.

Wireshark analysis confirmed the behavior of TCP SYN scanning, where SYN packets are sent and responses are analyzed to determine port status. The absence of SYN-ACK responses confirms that no services are openly accessible.

Overall, the network demonstrates a strong security posture with minimal exposure to threats.

Learning Outcomes

Understood port scanning techniques

Learned TCP SYN scan mechanism

Gained hands-on experience with Nmap

Learned packet analysis using Wireshark

Understood firewall behavior